

TRDP2: Limited-Angle 3D CT Reconstruction with R²-Gaussian

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Abstract

This pilot investigated how restricted angular acquisition affects 3D Gaussian CT reconstruction. Using one synthetic chest case, we compared 50 noise-free views over a full circle with 50 views over two differently oriented 120° intervals. Restricting coverage reduced final volume PSNR from 35.55 dB to 30.91 dB; rotating the limited acquisition by 90° reduced it further to 28.59 dB. The Gaussian pipeline substantially improved raw FDK reconstructions, but retained directional boundary errors. Held-out projection errors were close to the full-circle control inside each acquired interval and substantially larger outside it. Dominant error peaks shifted approximately with acquisition orientation. These findings motivate further study of geometry-aware reliability estimation, but do not establish generalization or an uncertainty-estimation method.

1 Objective and scope

The objective was to establish a reproducible 3D medical-imaging baseline and inspect its failures before designing a correction.

Three questions guided the pilot:

1. How much does restricting angular coverage degrade reconstruction when the number of training views is fixed?
2. Which errors are present in raw FDK, and which remain after the Gaussian reconstruction pipeline?
3. Do difficult viewing directions change when the acquired angular interval is rotated?

This report freezes the completed runs as exploratory pilot results. No projection-completion method, evidential model, or learned reliability estimator was trained.

2 Dataset and experimental methods

2.1 Data and acquisition

We used the synthetic cone-beam chest case `0_chest_cone` from the dataset distributed with R²-Gaussian [1, 2], originally under `cone_ntrain_50_angle_360`. The reference volume contains $256^3 = 16,777,216$ voxels. Reported intensities are dataset values, not calibrated Hounsfield units.

For the controlled experiments, projections were regenerated from this reference using TIGRE [3] and the supplied scanner geometry. Projection noise was disabled. Volume-axis ordering and detector orientation followed the repository’s generation conventions.

Table 1: Controlled acquisitions. Interval endpoints on the right are excluded. All cases use the same reference and scanner geometry.

Case	Acquired interval	Views	Spacing	Last view
Full-circle control	$[0^\circ, 360^\circ)$	50	7.2°	352.8°
Limited, start 0°	$[0^\circ, 120^\circ)$	50	2.4°	117.6°
Limited, start 90°	$[90^\circ, 210^\circ)$	50	2.4°	207.6°

The full-circle control is itself a sparse-view acquisition. View count is matched across cases, but angular spacing differs between the full-circle and limited-angle cases.

All cases share 100 held-out full-circle projections at

$$\theta_k = (k + 0.5) 3.6^\circ, \quad k = 0, \dots, 99.$$

These angles do not overlap any training grid. The rotated case uses exact copies of the reference volume and evaluation projections from the original controlled dataset.

2.2 Reconstruction and implementation

We used R²-Gaussian [1], starting from the authors’ official implementation [2]. Our working repository for the TRDP2 experiments is:

https://github.com/zunaaaaayed/r2_gaussian
Experiment branch: experiment/limited-angle-120

The upstream repository provides the baseline implementation, while our working repository is used for the controlled limited-angle experiments and diagnostic scripts.

Each acquisition received its own Feldkamp–Davis–Kress (FDK) [4] initialization, computed from its training projections using TIGRE. From the FDK volume, 50,000 locations with values above 0.05 were sampled, and their initial densities were multiplied by 0.15. Reconstruction then ran for 30,000 iterations with the same existing training settings, including total-variation regularization. Evaluation used the fixed final iteration, not a reference-selected best checkpoint.

Raw FDK volumes were separately regenerated and saved *before* thresholding, point sampling, and density rescaling. Their negative values were retained in numerical evaluation. Therefore, comparing raw FDK with final R² measures the combined effects of initialization conversion, representation, and optimization.

Experiments ran on an NVIDIA RTX 4000 Ada GPU with approximately 20 GB memory, Python 3.9.23, PyTorch 2.1.2 with CUDA 11.8, and TIGRE v2.3. An earlier initialization crash was resolved by removing a CUDA-context reset in the local TIGRE build; the generated initialization file remained exactly unchanged in the verification comparison.

2.3 Evaluation and consistency checks

Volume metrics included MAE, RMSE, signed bias, and PSNR with peak value 1:

$$\text{PSNR} = -10 \log_{10} \left(\frac{1}{N} \sum_{i=1}^N (\hat{x}_i - x_i)^2 \right).$$

Structural similarity (SSIM) [5] was computed using the repository’s evaluation implementation. The reported score aggregates slice-wise results across three array axes; it is not a single volumetric 3D-window SSIM calculation.

For angular diagnostics, saved voxel volumes were forward-projected with TIGRE and compared with the shared held-out reference projections. No independent intensity normalization or numerical clipping was applied. This evaluates voxel-volume projections, separately from the Gaussian renderer’s training loss.

Reprojecting the reference volume reproduced the stored evaluation projections exactly in the initial diagnostic: both MAE and maximum absolute error were zero. This establishes internal consistency with data generation, not independent validation of the physical scanner model.

3 Results and interpretation

3.1 Volume reconstruction quality

Table 2: Whole-volume results. Raw FDK was evaluated for the control and original limited-angle case.

Acquisition	Method	PSNR (dB)	SSIM	MAE	RMSE
360°	Raw FDK	24.17	–	0.04700	0.06188
360°	Final R ²	35.55	0.9419	0.00817	0.01670
120°, start 0°	Raw FDK	19.26	–	0.08610	0.10893
120°, start 0°	Final R ²	30.91	0.9010	0.01370	0.02849
120°, start 90°	Final R ²	28.59	0.8623	–	–

Restricting coverage to the original 120° interval reduced final PSNR by 4.64 dB relative to the control. Rotating this interval by 90° reduced PSNR by a further 2.31 dB, corresponding to approximately 1.70 times the voxel mean squared error of the original limited-angle result.

The Gaussian pipeline improved raw FDK PSNR by 11.38 dB for the control and 11.65 dB for the original limited-angle case. It substantially reduced artifacts, but did not remove the limited-angle accuracy deficit.

In raw FDK, 30.14% of control voxels and 35.23% of limited-angle voxels were negative. Both volumes contained more than eight million voxels above the initialization threshold, so neither lacked eligible sampling locations. Eligibility does not establish that a location represents accurately reconstructed tissue.

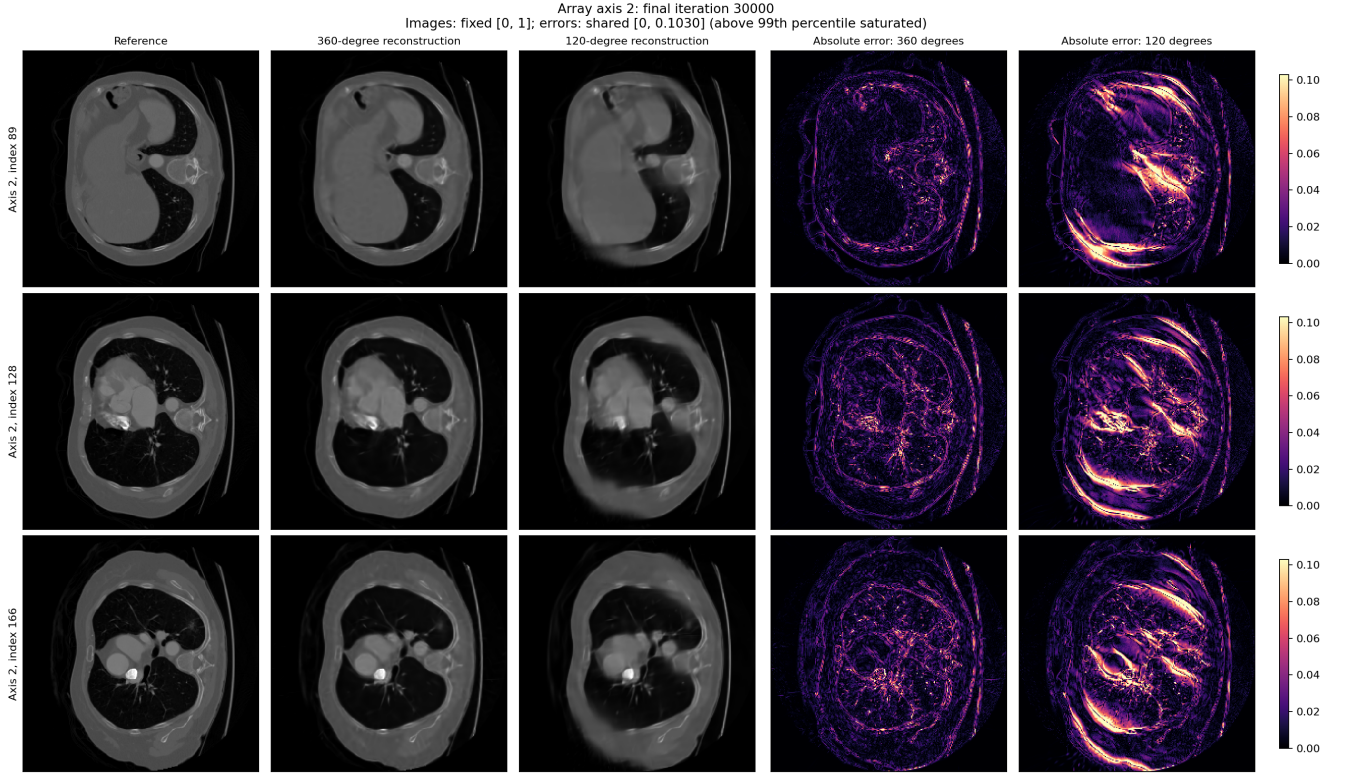


Figure 1: Reference, final control, and final original 120° reconstruction at three array-axis-2 slices, followed by absolute-error maps. The limited-angle result shows stronger directional smearing and boundary errors. Grayscale uses a fixed [0, 1] display window; error maps share the displayed scale, with high values saturated. Anatomical coordinate labels were not inferred from array axes.

Regional metrics also reveal cancellation hidden by overall bias. In the original 120° final volume, mean error was -0.00489 where reference intensity exceeded 0.05 and $+0.00367$ elsewhere, giving a near-zero overall bias. These intensity groups are exploratory masks, not anatomical segmentations.

3.2 FDK versus final projection accuracy

For the control, mean held-out projection MAE decreased from 0.01254 for FDK to 0.001170 for final R^2 . For the original limited-angle case, it decreased from 0.04330 to 0.002453, a 94.3% reduction.

Figure 2 shows that the improvement is not a uniform rescaling of the angular error pattern. Final limited-angle errors remain especially pronounced near 146° and 329°. Raw FDK also has broad elevated-error regions near these directions, but has additional substantial errors elsewhere.

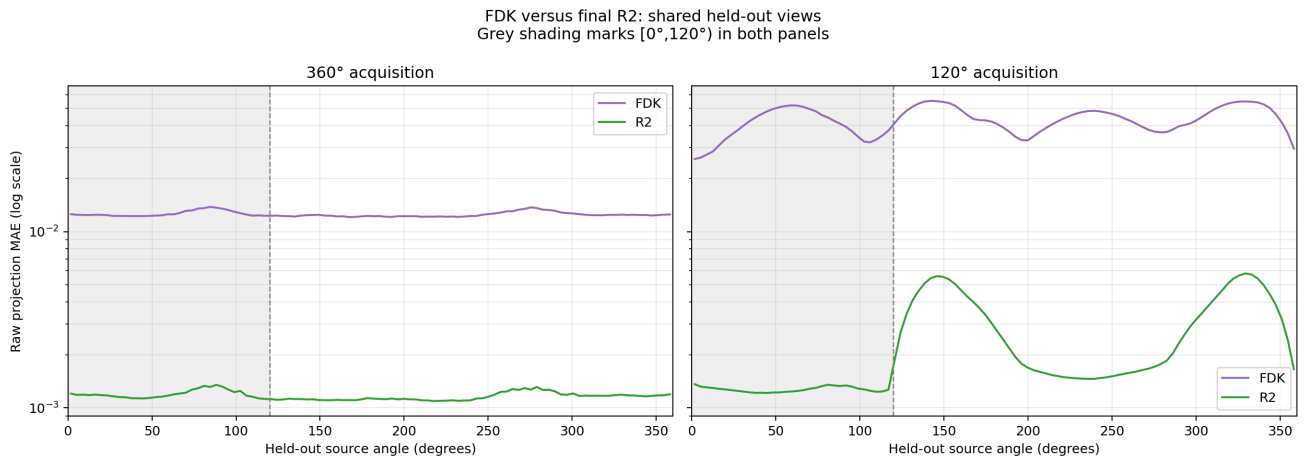


Figure 2: Held-out projection MAE for raw FDK and final R^2 , shown on a logarithmic vertical scale. The final reconstruction improves accuracy across directions, while retaining a direction-dependent residual pattern under limited coverage. Shading marks [0°, 120°] in both panels.

Similar peak locations do not prove that FDK initialization causes the final peaks: both stages use the same incomplete

measurements. An initialization comparison would be needed to separate these effects.

3.3 Effect of rotating the acquisition

Table 3: Final held-out projection errors. Ratios compare each case with the control at the same evaluation angles. Both limited acquisitions have 33 inside and 67 outside views.

Acquisition	Sector	MAE	RMSE	MAE/control
Start 0°	Inside	0.001272	0.002350	1.061
Start 0°	Outside	0.003034	0.006639	2.627
Start 90°	Inside	0.001195	0.002327	1.050
Start 90°	Outside	0.003692	0.008643	3.114

Both limited acquisitions predict held-out views inside their acquired intervals nearly as accurately as the control. However, errors outside those intervals are substantially larger. The rotated acquisition has slightly lower inside-arc projection MAE despite worse volume quality. Thus, good agreement within the acquired arc is insufficient to establish accurate reconstruction.

To compare error patterns, we used

$$\theta_{\text{relative}} = (\theta_{\text{absolute}} - \theta_{\text{start}}) \bmod 360^\circ.$$

The dominant absolute-angle peaks moved from 145.8° and 329.4° to 243.0° and 66.6°, respectively. Each shift is 97.2° around the circle.

In relative coordinates, the peaks occur at 145.8° and 329.4° for the original case, and 153.0° and 336.6° for the rotated case. The residual difference is 7.2°, or two evaluation grid steps. The alignment is approximate; peak magnitude and curve shape also change.

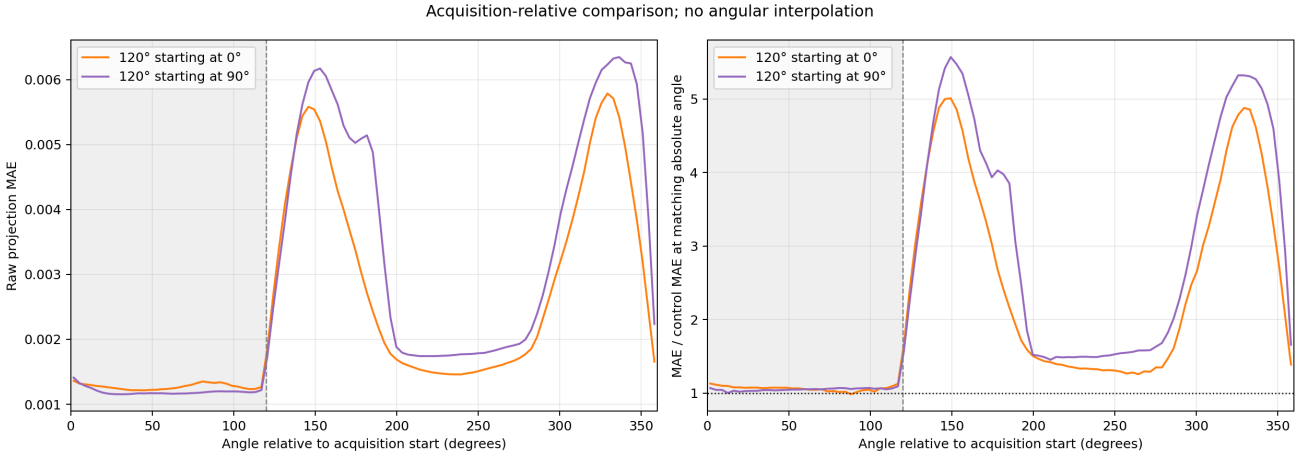


Figure 3: Acquisition-relative projection errors. Left: raw MAE. Right: MAE divided by control MAE at the matching absolute direction. Similar peak locations support an acquisition-dependent error pattern, while differences in height and shape show that orientation effects are not captured by a simple shift alone.

4 Limitations and next steps

This pilot uses one chest volume, noise-free synthetic measurements, one limited angular span, two orientations, and one training run per configuration. Its findings do not establish statistical robustness or generalization to other anatomies, acquisition spans, noise levels, or real CT data.

The reference volume and held-out projections provide evaluation targets. Error maps and angular error curves computed from these targets are *not* uncertainty estimates available at inference time. The full-circle reconstruction used to form diagnostic ratios would likewise not normally be available for a limited-angle patient scan.

The current results motivate a focused next question:

Can acquisition geometry and reconstruction-derived information predict the reliability of missing-view projections without access to reference measurements at inference time?

Next week’s work should define that prediction target and its evaluation protocol, compare simple geometry-based and reconstruction-dependent baselines, and reserve additional acquisitions or volumes for evaluation. Evidential learning should be considered as a method to test against these baselines, rather than assumed to be superior.

5 Pilot status and reproducibility

The reported runs are frozen as pilot results, with evaluation fixed at iteration 30,000. Scripts, small diagnostic tables, figures, input hashes, and protocol records should be retained on the experiment branch. Large datasets, volumes, and checkpoints remain on the university computer. No claim is made that the entire experiment is archived on GitHub until the latest files have been committed and pushed.

The principal finding is that the Gaussian pipeline markedly improves its FDK starting point, while limited-angle errors remain structured and approximately tied to acquisition orientation. This provides an empirical basis for the next reliability study.

References

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